Enthalpy and Calorimetry

Answer key with worked solutions

A. Understanding

1. Classify.

> (a) **exothermic**, ΔH negative (b) **endothermic**, ΔH positive (heat is absorbed to melt the ice) (c) **endothermic**, ΔH positive (the solution cools because heat is absorbed) (d) **exothermic**, ΔH negative

2. Thermochemical equations.

> (a) 2 × (−890.5 kJ) = **−1781.0 kJ** (b) **+890.5 kJ** (c) $n = \dfrac{8.02\ \text{g}}{16.04\ \text{g/mol}} = 0.500\ \text{mol}$; heat released = 0.500 mol × 890.5 kJ/mol = **445 kJ** (ΔH = −445 kJ)

B. q = mcΔT

3. Copper.

> $q = (40.0\ \text{g})(0.385\ \text{J/(g·°C)})(75.0\ ^\circ\text{C} - 25.0\ ^\circ\text{C}) =$ **770. J** (g and °C cancel)

4. Specific heat.

> $c = \dfrac{q}{m\Delta T} = \dfrac{900.\ \text{J}}{(100.\ \text{g})(20.0\ ^\circ\text{C})} =$ **0.450 J/(g·°C)** (close to iron, 0.449 J/(g·°C))

5. Final temperature.

> $$\Delta T = \frac{q}{mc} = \frac{5.00 \times 10^3\ \text{J}}{(150.\ \text{g})(4.184\ \text{J/(g·°C)})} = 7.97\ ^\circ\text{C}$$
>
> $T_{\text{final}} = 22.0\ ^\circ\text{C} + 7.97\ ^\circ\text{C} =$ **30.0 °C**

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C. Calorimetry

6. Ammonium nitrate.

> $\Delta T = 16.0\ ^\circ\text{C} - 22.0\ ^\circ\text{C} = -6.0\ ^\circ\text{C}$
> $q_{\text{soln}} = (108.0\ \text{g})(4.184\ \text{J/(g·°C)})(-6.0\ ^\circ\text{C}) = -2711\ \text{J}$, so $q_{\text{reaction}} = +2711\ \text{J}$
> $n = \dfrac{8.00\ \text{g}}{80.04\ \text{g/mol}} = 0.09995\ \text{mol}$
> $\Delta H = \dfrac{+2711\ \text{J}}{0.09995\ \text{mol}} = +2.7 \times 10^4\ \text{J/mol} =$ **+27 kJ/mol** (endothermic; 2 significant figures from ΔT)

7. Neutralization.

> Mass of solution = 50.0 mL × 1.00 g/mL = 50.0 g; $\Delta T = 33.5\ ^\circ\text{C} - 20.0\ ^\circ\text{C} = 13.5\ ^\circ\text{C}$
> $q_{\text{soln}} = (50.0\ \text{g})(4.184\ \text{J/(g·°C)})(13.5\ ^\circ\text{C}) = 2824\ \text{J}$, so $q_{\text{reaction}} = -2824\ \text{J}$
> $n(\text{H}_2\text{O}) = 2.00\ \text{mol/L} \times 0.0250\ \text{L} = 0.0500\ \text{mol}$
> $\Delta H = \dfrac{-2824\ \text{J}}{0.0500\ \text{mol}} = -5.65 \times 10^4\ \text{J/mol} =$ **−56.5 kJ/mol**

D. Challenge

8. Ethanol under a can.

> (a) Heat released: $\dfrac{0.500\ \text{g}}{46.07\ \text{g/mol}} \times 1366.8\ \text{kJ/mol} = 14.83\ \text{kJ}$
>
> $$\Delta T_{\text{max}} = \frac{1.483 \times 10^4\ \text{J}}{(200.\ \text{g})(4.184\ \text{J/(g·°C)})} = 17.7\ ^\circ\text{C}$$
>
> (b) Percentage transferred = $\dfrac{12.4\ ^\circ\text{C}}{17.73\ ^\circ\text{C}} \times 100\% =$ **69.9%** (the rest heated the can and the air)

9. Iron in water.

> Heat lost by iron = heat gained by water:
> $(50.0\ \text{g})(0.449\ \text{J/(g·°C)})(100.0\ ^\circ\text{C} - T) = (100.\ \text{g})(4.184\ \text{J/(g·°C)})(T - 20.0\ ^\circ\text{C})$
> $22.45\ \text{J/°C} \times (100.0\ ^\circ\text{C} - T) = 418.4\ \text{J/°C} \times (T - 20.0\ ^\circ\text{C})$
> $2245\ \text{J} + 8368\ \text{J} = (418.4 + 22.45)\ \text{J/°C} \times T$, so $T = \dfrac{10613\ \text{J}}{440.85\ \text{J/°C}} =$ **24.1 °C**

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